CONCEPT 1 — TERRACOTTA COURTYARD HOUSE

Two storeys in the ONT band between 3-storey neighbours · 1.5 m west service passage · east light court · shaded north balcony



Blender render — street view (golden hour)

A two-storey family house in northern Vietnamese timber, brick and terracotta, kept wholly inside the 13.5 × 19.7 m road-side ONT band. The entrance hall (Huyền Quan) is on the road. Living space runs north towards the garden. A 2.5 × 4 m east light court brings daylight and air into the middle of the deep plan. A 1.5 m service passage on the west side serves the wet kitchen and laundry without crossing the living rooms. Upstairs, three bedrooms, an altar room and a family lounge open onto a 2.2 m shaded balcony that looks over the existing garden land towards the river. It is designed for a Vietnamese street, with a 3-storey house assumed on each side. The east wall is blank because it faces the neighbour's party wall 0.6 m away. The west wall has only high-level windows onto the service passage. Daylight comes from the street, the light court and the garden. The only structure on the DNN land is the pair of boundary walls running down to the water.



Concept 1 brief image (inspiration)

THE CONCEPT IN NUMBERS

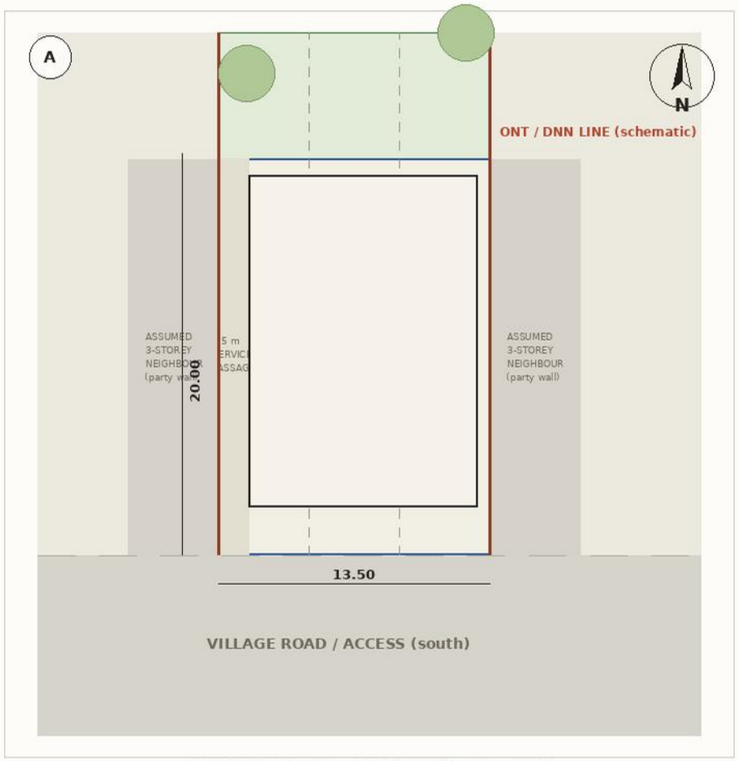
Footprint	188.1 m² (71% of ONT band)
Ground floor	178 m² enclosed
Upper floor	153 m² enclosed + 25 m² balcony
Total enclosed	331 m²
Constraint check	PASS — all structures in ONT; DNN planting only

Walkthrough test: 24/24 rooms reachable

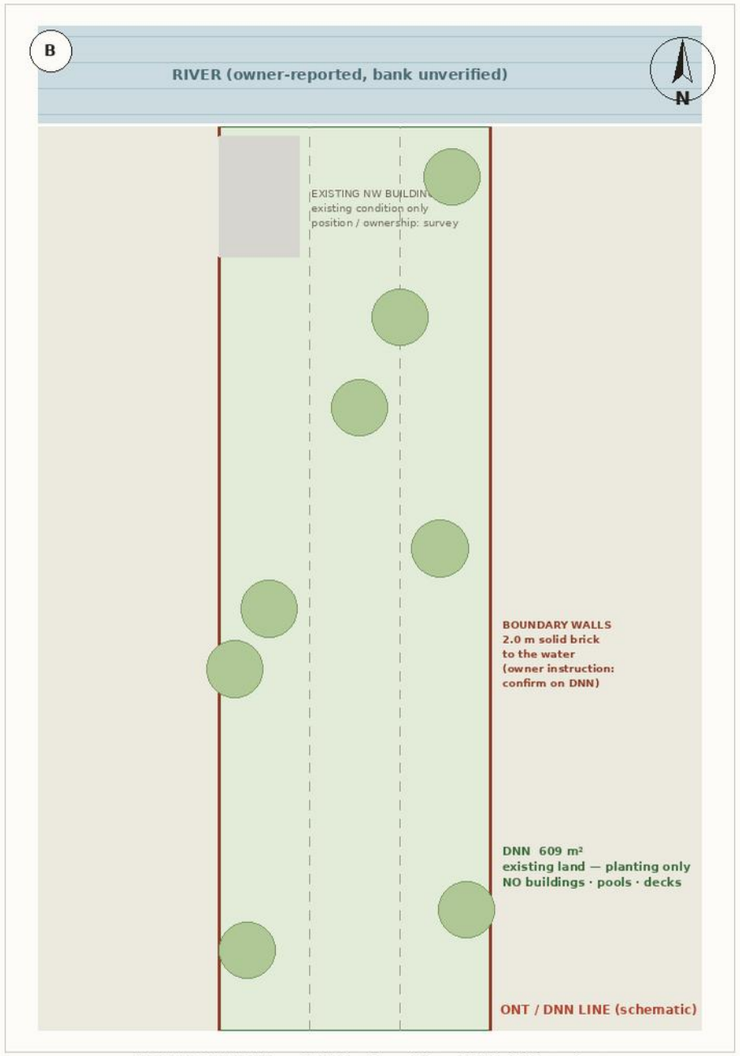
The concept image asked for about 230–250 m² on the ground floor. The ONT band is only about 266 m², so the plan is fitted inside it with a 1.5 m west service passage, a 2.4 m forecourt and 0.6 m eaves. That gives 178 m² enclosed on the ground floor. The house sits on the east boundary, which needs a boundary/setback check.

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

SITE PLAN · 1:200 at A3 · shown in two parts so the narrow 13.5 m plot stays readable



SOUTHERLY PART — ONT band, house, road



NORTHERLY PART — DNN land, existing NW building, river

NOTES

Schematic site from owner information only: three side-by-side 4.5 m titles, about 13.5 × 65 m. No survey. Corners, levels, bank edge and road reserve are unverified.

Structures are shown ONLY inside the ONT road-side band (3 × 88.8 = 266.4 m²). The 609 m² DNN band is existing land, shown as planting only.

The title sketch shows 6.5 m to the road centreline. It is NOT treated as a verified setback.

Assumed: a 3-storey house on each side built to the property line to about the same 20 m depth. So the east wall is blank (party wall) and the west wall has high-level windows only.

Side boundary walls run the full length to the water on both property lines (owner instruction). This is the only structure on DNN land. Confirm it is permitted.

The existing north-west building is an existing condition only. Its position and ownership are to be surveyed. It is not altered.

Subject to legal merger of the three ONT sections, setbacks, river corridor and permits.

KEY

- ONT — buildable band
- DNN — land / planting only
- service passage
- existing NW building
- assumed 3-storey neighbours
- boundary walls
- house footprint
- assumed / context

0 10 20 30 40 50 metres (assumed — no survey)

CONCEPT 1 — TERRACOTTA COURTYARD HOUSE

Two storeys in the ONT band between 3-storey neighbours · 1.5 m west service passage · east light court · shaded north bale

SCALE	SHEET	STATUS
1:200 at A3	**C1-01** site plan	**FOR REVIEW** *not for construction*

enclosed 178.1 m² · level 1 of 2

GROUND FLOOR · FFL +0.45 / +3.75



enclosed 153.0 m² · level 2 of 2

UPPER FLOOR · FFL +3.75 / +6.85



0 10 20 30 40 50 metres (assumed — no survey)

KEY

- HABITABLE
- SERVICE / WET
- CIRCULATION
- STORAGE
- SPECIAL
- OUTDOOR
- PLANT / VOID

NOTES

CONCEPT ONLY — NOT FOR CONSTRUCTION. Dimensions are taken from owner-supplied title information; no survey. Subject to survey, legal merger, setbacks and permits.

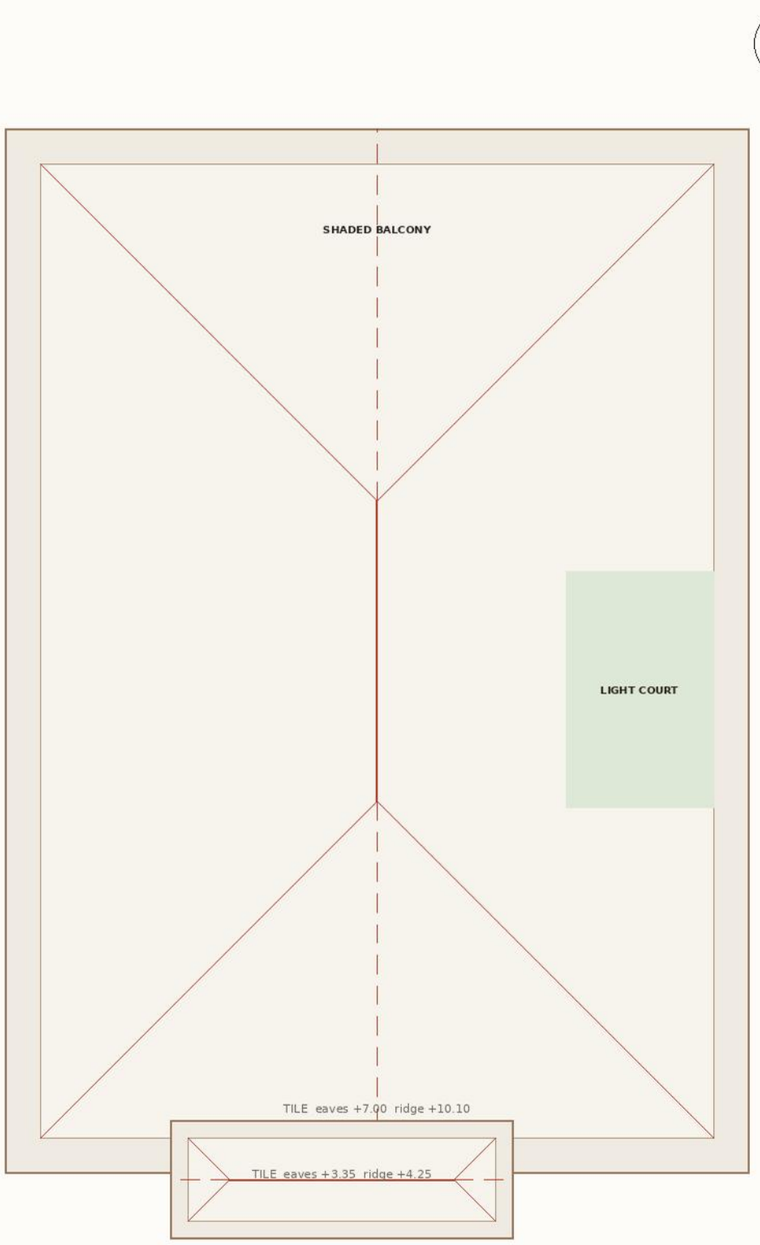
Areas are measured from the room rectangles; the balcony and light court are excluded from enclosed area.

CONCEPT 1 — TERRACOTTA COURTYARD HOUSE

Two storeys in the ONT band between 3-storey neighbours · 1.5 m west service passage · east light court · shaded north balc

SCALE	SHEET	STATUS
1:100 at A3	C1-02	FOR REVIEW
	floor plans 1–2	not for construction

ROOF PLAN · looking down



0 10 20 30 40 50 metres (assumed — no survey)

KEY

- roof covering
- ridge / hip
- fascia / eaves
- open slot below

ROOF KEY

- concrete tile on timber — eaves +7.00, ridge +10.10
- ridge / hip line
- fascia / eaves edge, 1.05 m overhang
- rain-garden slot, open to the sky
- outline of the level below (hidden line)

NOTES

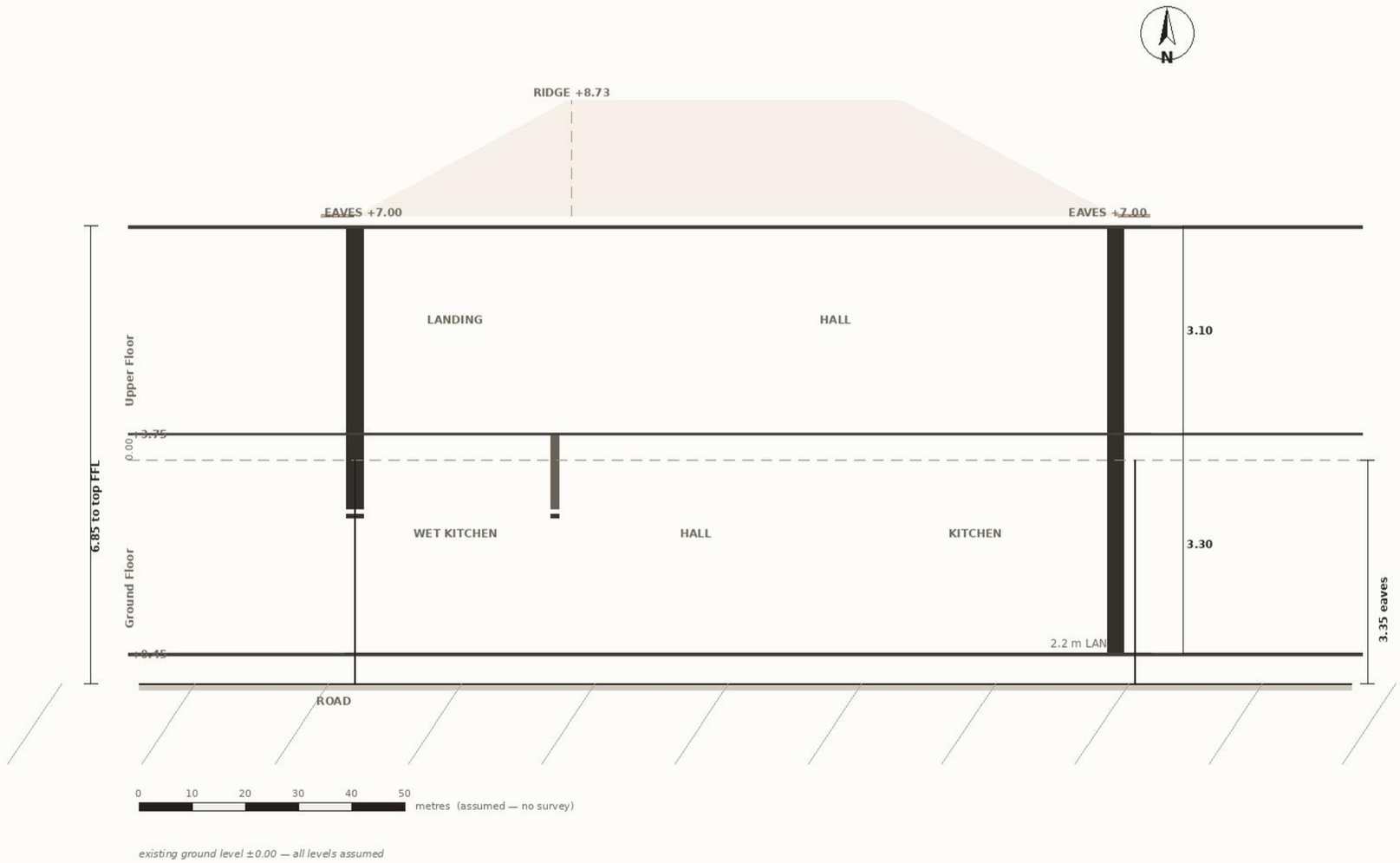
Ridge and eaves heights come from the option massing: eaves +7.00 m, ridge +10.10 m above datum, 0.60 m overhang all round.
Hips, ridges and the rain-garden slot are derived from the same rectangles as the plans and the 3D model.
Rainwater from the slot and the roof gardens is the only visible drainage; confirm the neighbour-boundary falls before detailing.

CONCEPT 1 — TERRACOTTA COURTYARD HOUSE

Two storeys in the ONT band between 3-storey neighbours · 1.5 m west service passage · east light court · shaded north balc

SCALE	SHEET	STATUS
1:100 at A3	**C1-03** roof plan	**FOR REVIEW** *not for construction*

SECTION A–A · looking north at y = 7.00 m (assumed cut)



existing ground level ±0.00 — all levels assumed

KEY

- cut external wall, 0.30
- cut internal wall, 0.15
- glazing in the cut
- roof build-up in section

NOTES

Datum 0.00 = existing road and ground level. Finished ground floor +0.45 m, first +3.75 m, eaves +3.35 m, ridge +10.10 m.

Slabs, walls and levels come from the same rectangles as the plans, so the section and the plans can never disagree.

The 2.2 m rain-garden slot is open to the sky on every level: draw it as a full-height void when the structure is detailed.

The cut is at y = 7.00 m. Move it on the plans to suit a later survey.

CONCEPT 1 — TERRACOTTA COURTYARD HOUSE

Two storeys in the ONT band between 3-storey neighbours · 1.5 m west service passage · east light court · shaded north bale

SCALE	SHEET	STATUS
1:100 at A3	**C1-04** section A–A	**FOR REVIEW** *not for construction*

AREA SCHEDULE · net internal, measured from the plan rectangles

Room	Type	Area m²	Cum. m²
Ground Floor · FFL +0.45			
Dining	Habitable	23.6	23.6
Robe	Habitable	5.8	29.4
Living (garden view)	Habitable	33.7	63.0
Master Suite	Habitable	21.6	84.6
Light Court	Plant / Void	10.0	94.6
Ensuite	Service / Wet	5.8	100.4
Gallery	Circulation	11.9	112.3
Entry (Huyền Quan)	Circulation	10.1	122.3
Hall	Circulation	13.4	135.8
Kitchen	Service / Wet	13.4	149.2
Laundry / WC	Service / Wet	7.2	156.4
Pantry / Store	Storage	10.1	166.5
Stair	Circulation	12.0	178.5
Wet Kitchen	Service / Wet	9.6	188.1
Level total, enclosed		**178.1**	
Upper Floor · FFL +3.75			
Bedroom 1	Habitable	16.0	16.0
Bedroom 2	Habitable	16.0	31.9
Bedroom 3	Habitable	19.7	51.7
Family Lounge	Habitable	19.7	71.4
Altar Room	Special	14.1	85.5
Shaded Balcony	Outdoor	25.1	110.6
Bathroom	Service / Wet	7.2	117.8
Bathroom 2	Service / Wet	7.8	125.6
Light Court	Plant / Void	10.0	135.6
Hall	Circulation	15.1	150.7
Gallery	Circulation	11.6	162.3
Landing	Circulation	9.6	171.9
Linen	Storage	4.2	176.1
Stair	Circulation	12.0	188.1
Level total, enclosed		**153.0**	

AREA SUMMARY	Area
Level 1 — Ground Floor (enclosed)	178.1 m²
Level 2 — Upper Floor (enclosed)	153.0 m²
Total enclosed floor area	**331.1 m²**

ORDER OF COST — TURNKEY, NORTHERN VIET NAM

INDICATIVE BUILD COST	M VND	+10% prov.
Basic build — 4.9 M VND/m²	1 622	1 785
Mid build — 6.5 M VND/m²	2 152	2 368
High build — 8.5 M VND/m²	2 815	3 096

NOTES

Areas are net internal floor areas computed from the room rectangles. Terraces, loggias, the rain-garden slot, the plant deck and roof gardens are excluded from the totals but shown on the plans.

The build-cost range is a rule of thumb from published Vietnamese cost guides for northern Viet Nam, not a quotation. It excludes site works, boundary treatment and furniture.

Confirm with a local quantity surveyor and at least two builders before committing to a budget.

NOTES

No habitable room here is below 2.7 m clear. The ground floor is a complete one-level home — bedroom, bathroom, living, dining and kitchen all at +0.45 — so the family can live on one level for as long as they want to.

Order-of-cost at 4.9–8.5 M VND/m² turnkey (northern Viet Nam) is in the table on the left. It is a rule of thumb, not a quotation.

ROOM TYPE KEY

- HABITABLE
- SERVICE / WET
- CIRCULATION
- STORAGE
- SPECIAL
- OUTDOOR
- PLANT / VOID

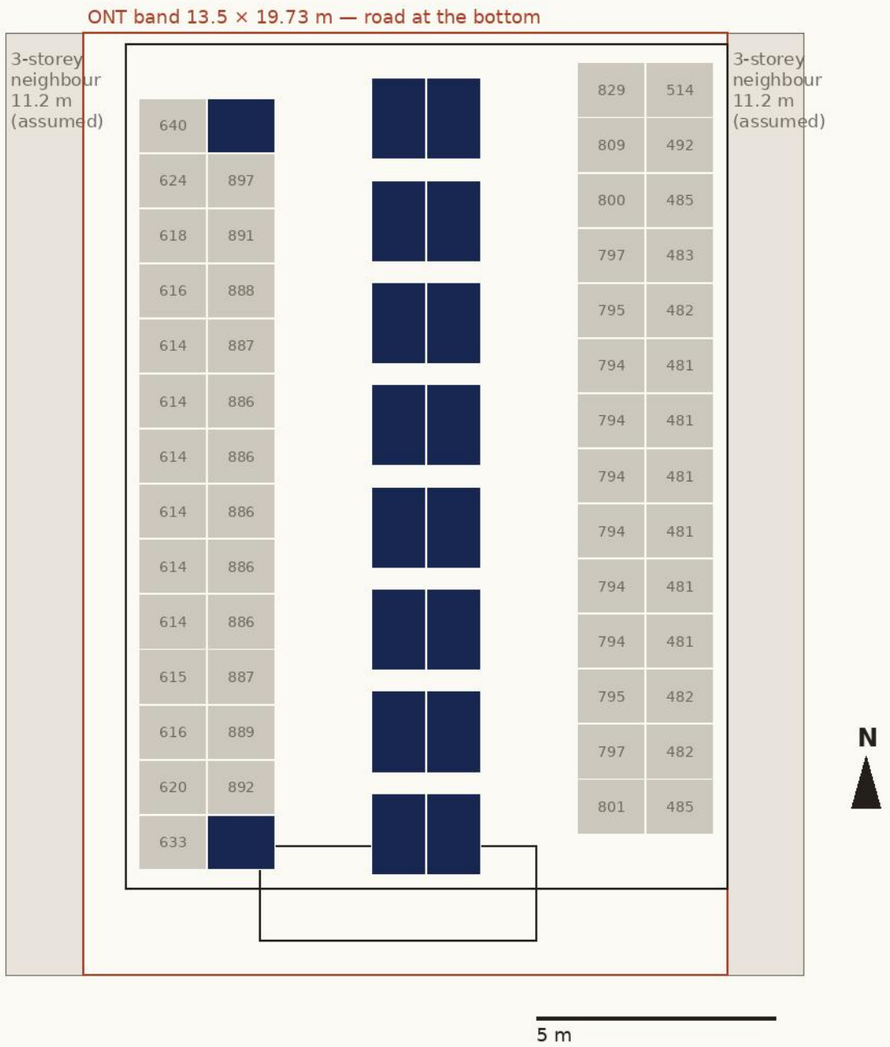
CONCEPT 1 — TERRACOTTA COURTYARD HOUSE

Two storeys in the ONT band between 3-storey neighbours · 1.5 m west service passage · east light court · shaded north balc

SCALE	SHEET	STATUS
as noted	**C1-05** area schedule	**FOR REVIEW** *not for construction*

Rooftop solar — energy per season

18 × 440 Wp panels = 7.9 kWp on the roof · latitude 21°N (rounded) · PVGIS 5.3 TMY (ERA5), typical year · shaded by the assumed 3-storey neighbours



Recommended panels

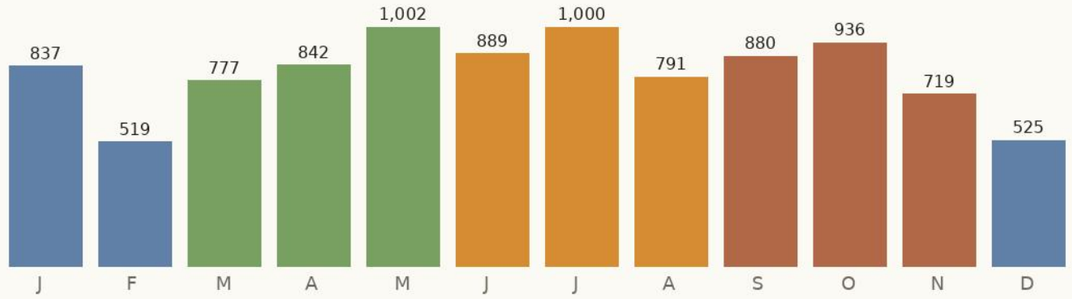
Fit but not worth it: number = kWh per kWp a year (an unshaded ideal panel gives 1281)

Roof surface	Tilt / faces	Fit / used	kWh/kWp	no neighbours
Main roof west slope	35° W	28 / 2	755	1134
Main roof flat top	10° S	16 / 16	1267	1270
Main roof east slope	35° E	28 / 0	642	1089

ENERGY FROM THE RECOMMENDED ARRAY

Season	Months	kWh	kWh / day	Share
Winter	Dec–Feb	1,881	20.9	19 %
Spring	Mar–May	2,621	28.5	27 %
Summer	Jun–Aug	2,680	29.1	28 %
Autumn	Sep–Nov	2,535	27.9	26 %
Year		**9,718**	**26.6**	

Without the assumed neighbours the same panels would give 9,939 kWh a year (+2 %). All 72 panels that physically fit would give 26,129 kWh, but the extra ones are on shaded or east/west/north slopes.



IS IT WORTH IT? (indicative)

	Recommended	Best-value size
Panels / kWp	18 / 7.9	7 / 3.1
Installed cost	111 M VND	43 M VND
Bill saving, first year	9 M VND	8 M VND
Output used in the house	26 %	54 %
Share of the bill-kWh covered	36 %	31 %
Simple payback	14 years	6 years
Net over 25 years	76 M VND	128 M VND

Assumes: an example air-conditioned family using about 6,890 kWh a year (bill 21 M VND a year), EVN household tiers from 1 Nov 2026 + 8 % VAT, 14 M VND per kWp installed, nothing paid for power sent to the grid (the buy-back rate is not yet published), 0.5 %/yr panel ageing, 1 %/yr upkeep. Surplus power is the limit: the house uses most of its power in the evening. Try other numbers at ali3b.github.io/home-design/solar.html.

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Batteries and living off-grid

Hour-by-hour over a typical year · battery 5 M VND per usable kWh incl. hybrid inverter, replaced once (year 13, 60 %) · nothing paid for surplus power

WITHOUT AIR-CON — example household using about 3,600 kWh a year (bill 9 M VND a year)

Panels	Battery	Cost	From sun: W Sp Su Au	Year	No-grid days	Saving / yr	Payback	Net over 25 years
7 (3.1 kWp)	none	43 M VND	30 36 38 35	35 %	0	4 M VND	13 yrs	37 M VND
7 (3.1 kWp)	**5 kWh**	**68 M VND**	63 77 78 75	**73 %**	**0**	**7 M VND**	**10 yrs**	**75 M VND ← best value**
7 (3.1 kWp)	10 kWh	93 M VND	76 93 93 93	89 %	248	8 M VND	16 yrs	63 M VND
7 (3.1 kWp)	15 kWh	118 M VND	77 95 96 94	90 %	275	8 M VND	21 yrs	26 M VND
18 (7.9 kWp)	none	111 M VND	34 42 45 38	40 %	0	4 M VND	> 25 yrs	-37 M VND
18 (7.9 kWp)	5 kWh	136 M VND	75 85 88 82	83 %	0	8 M VND	24 yrs	7 M VND
18 (7.9 kWp)	10 kWh	161 M VND	96 100 100 100	99 %	353	9 M VND	> 25 yrs	-4 M VND
18 (7.9 kWp)	15 kWh	186 M VND	98 100 100 100	99 %	359	9 M VND	> 25 yrs	-43 M VND

WITH AIR-CON — example household using about 6,890 kWh a year (bill 21 M VND a year)

Panels	Battery	Cost	From sun: W Sp Su Au	Year	No-grid days	Saving / yr	Payback	Net over 25 years
7 (3.1 kWp)	none	43 M VND	29 33 30 31	31 %	0	8 M VND	6 yrs	128 M VND
7 (3.1 kWp)	5 kWh	68 M VND	56 53 39 52	48 %	0	12 M VND	6 yrs	179 M VND
7 (3.1 kWp)	10 kWh	93 M VND	67 61 40 61	54 %	27	13 M VND	8 yrs	164 M VND
7 (3.1 kWp)	15 kWh	118 M VND	67 61 40 61	54 %	30	13 M VND	10 yrs	124 M VND
18 (7.9 kWp)	none	111 M VND	34 38 37 36	36 %	0	9 M VND	14 yrs	76 M VND
18 (7.9 kWp)	5 kWh	136 M VND	68 62 52 60	59 %	0	14 M VND	11 yrs	149 M VND
18 (7.9 kWp)	10 kWh	161 M VND	93 83 65 83	78 %	116	17 M VND	10 yrs	191 M VND
18 (7.9 kWp)	**15 kWh**	**186 M VND**	95 94 76 94	**88 %**	**192**	**19 M VND**	**11 yrs**	**192 M VND ← best value**

CAN THE HOUSE LIVE OFF-GRID?

Without air-con: YES in a typical year with 18 panels (7.9 kWp) and a 40 kWh battery, about 311 M VND. That costs 242 M VND more over 25 years than staying on the grid.

Nearly: 18 panels with 10 kWh covers all but 34 kWh a year (12 short days, the worst needing 5.0 kWh).

With air-con: YES in a typical year with 72 panels (31.7 kWp, every panel that fits, including the shaded slopes) and a 40 kWh battery, about 644 M VND. That costs 378 M VND more over 25 years than staying on the grid.

Nearly: 72 panels with 20 kWh covers all but 8 kWh a year (6 short days, the worst needing 3.8 kWh).

A "typical year" leaves out the worst years. Northern winters bring weeks of drizzle and low cloud, so an off-grid house needs a margin: about 30 % more battery and panels, or a small generator for bad spells. Off-grid also wastes the summer surplus. Staying connected with a 5–10 kWh battery gets most of the benefit for far less money, and keeps the lights on in power cuts.

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Blender views

Cycles renders from the same data as the drawings and the walkthrough



Street view



From the garden (DNN land, planting only)

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Blender views

Cycles renders from the same data as the drawings and the walkthrough



From the river end, between the boundary walls

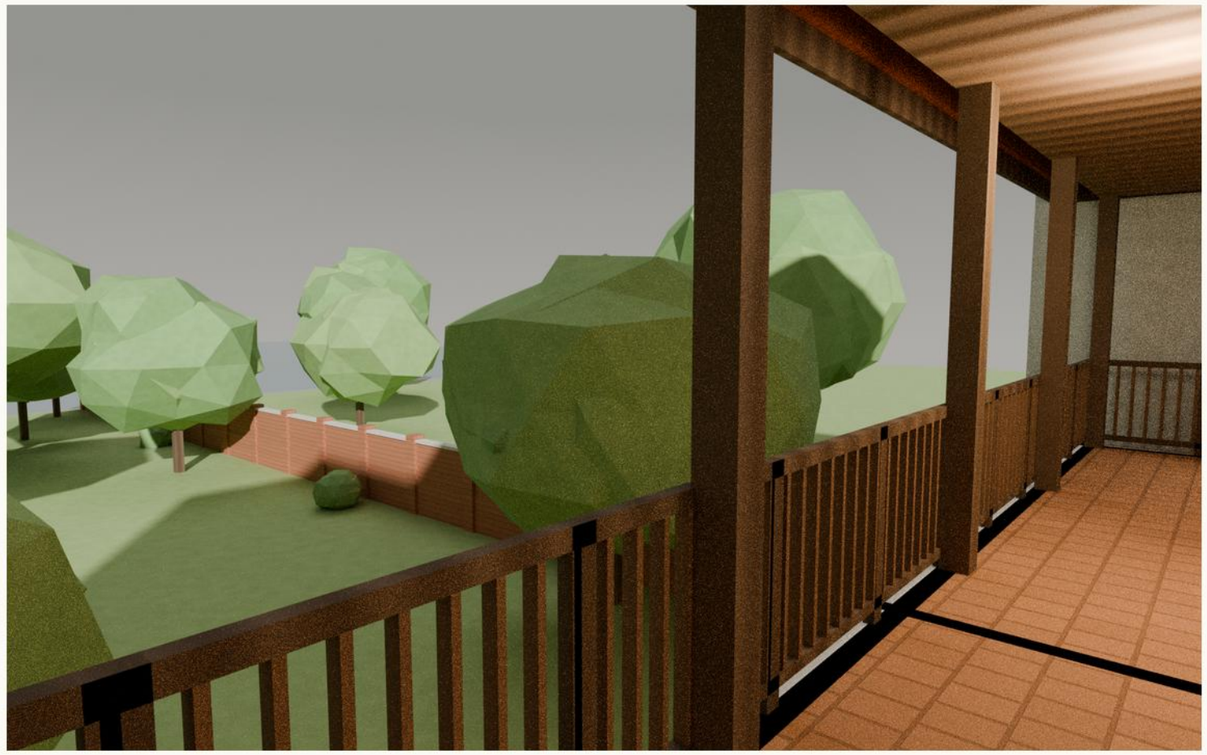


Aerial

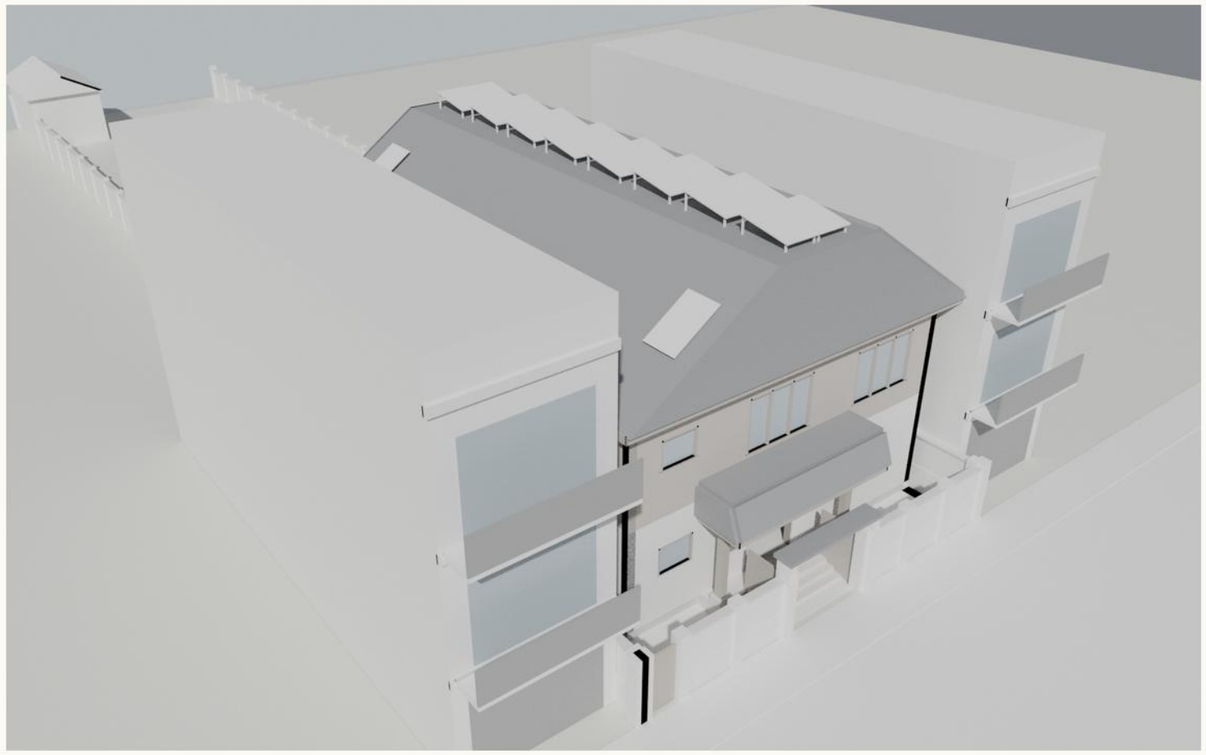
CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Blender views

Cycles renders from the same data as the drawings and the walkthrough



Upper balcony looking north



Clay massing

Walkthrough (Unity)

Interactive model: these are fixed viewpoints you can jump to with keys 1–9



Street



Entry



Living



Kitchen

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Walkthrough (Unity)

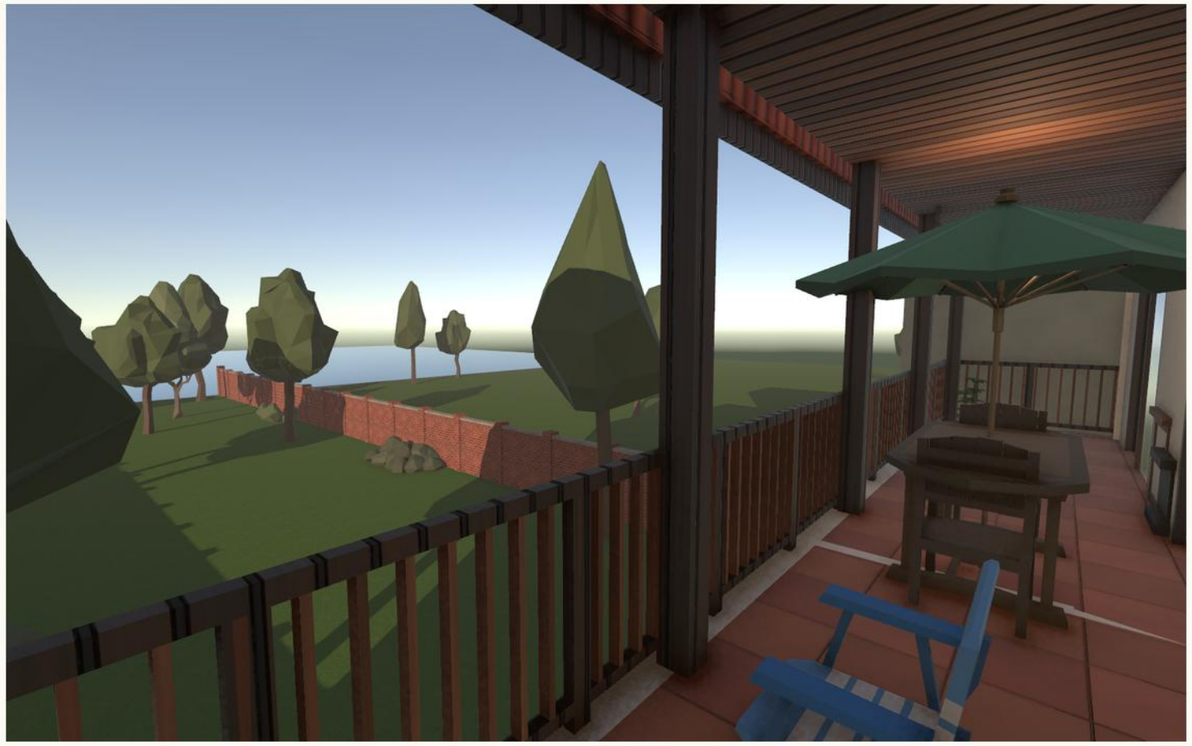
Interactive model: these are fixed viewpoints you can jump to with keys 1–9



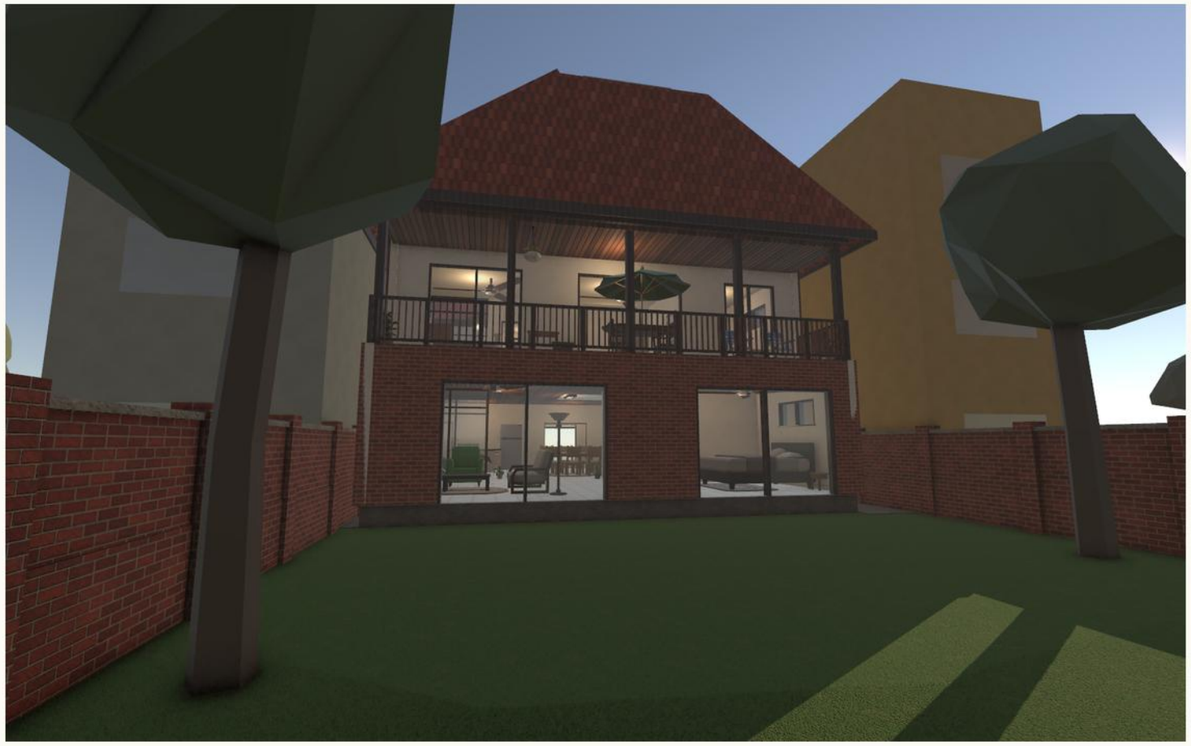
Master



Lounge



Balcony



Garden

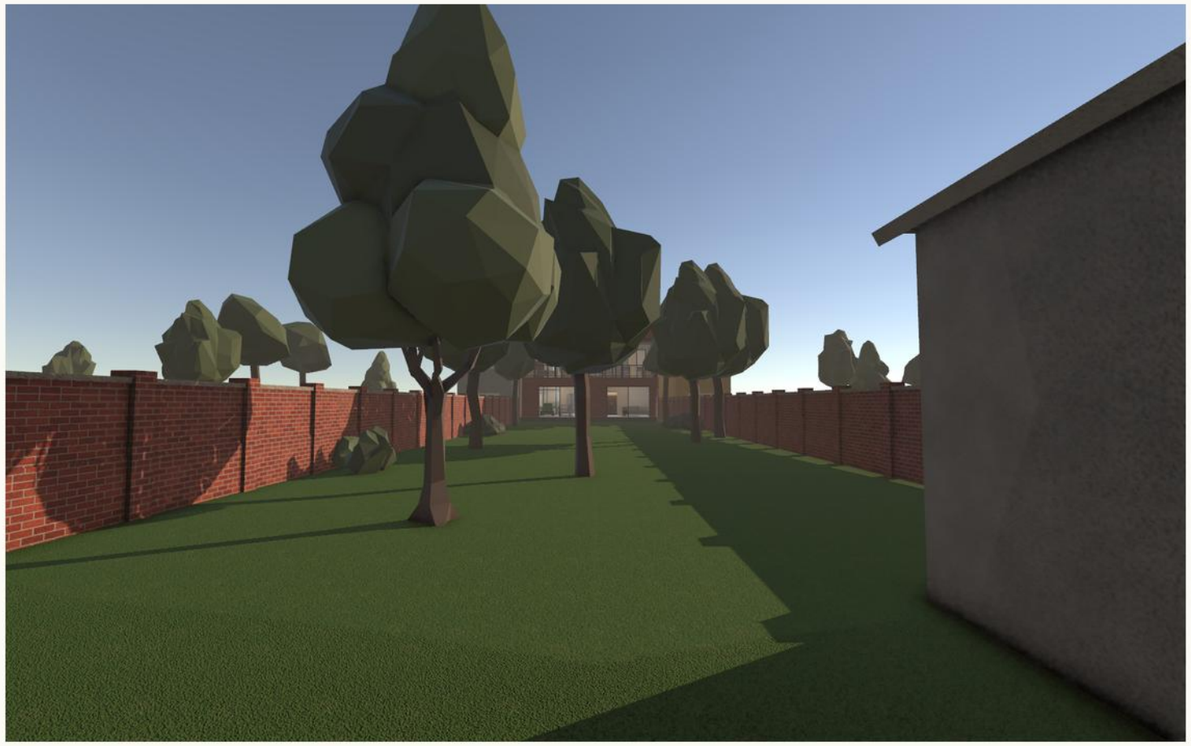
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Walkthrough (Unity)

Interactive model: these are fixed viewpoints you can jump to with keys 1–9



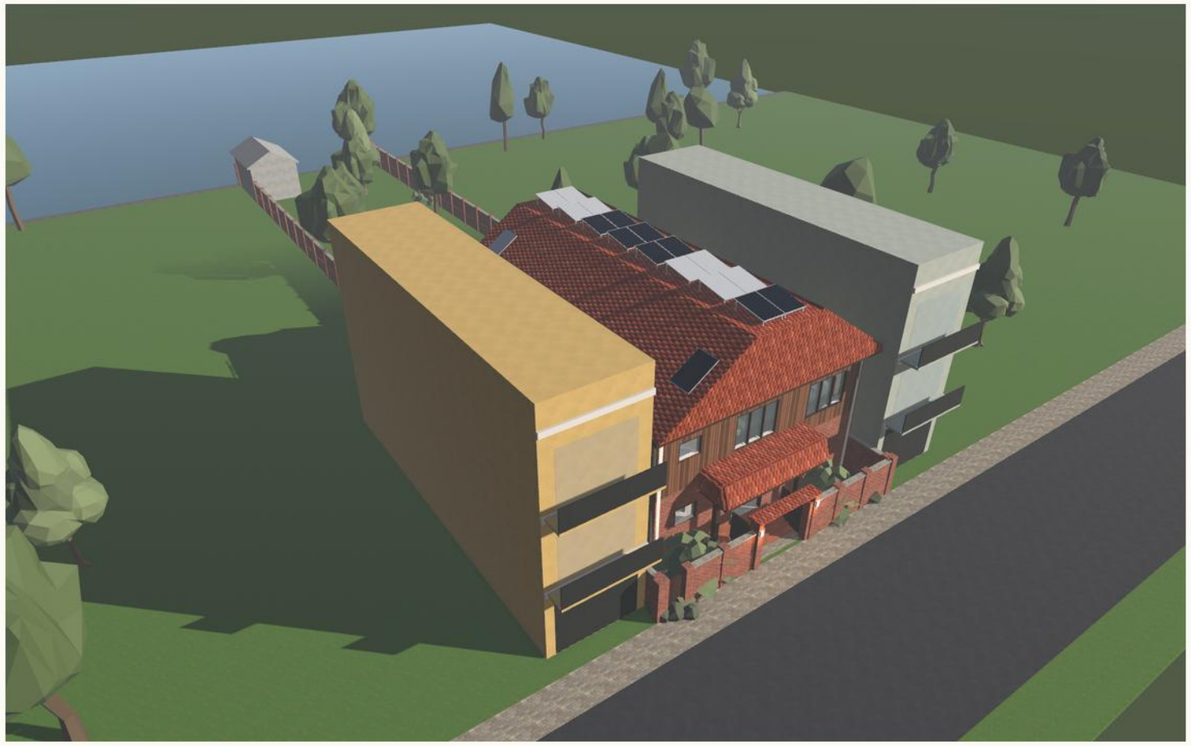
Aerial



Riverside



Dollhouse Ground

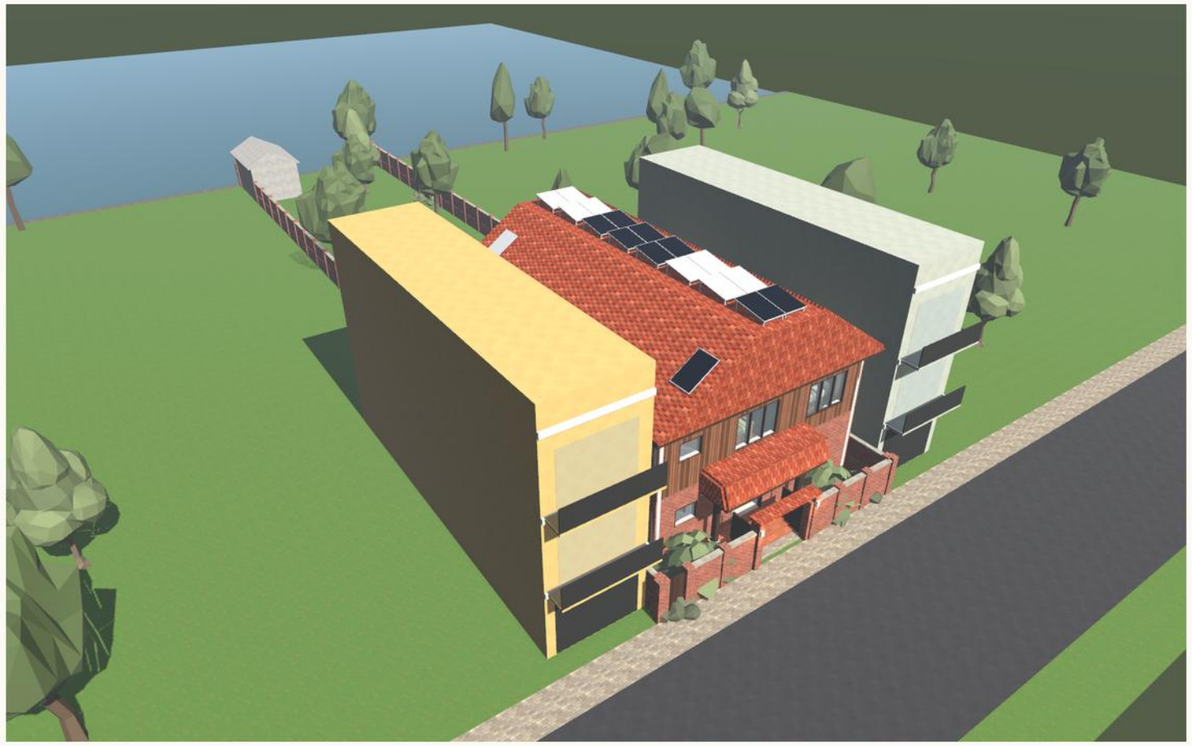


Sun study: winter solstice, 09:00 solar time

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Walkthrough (Unity)

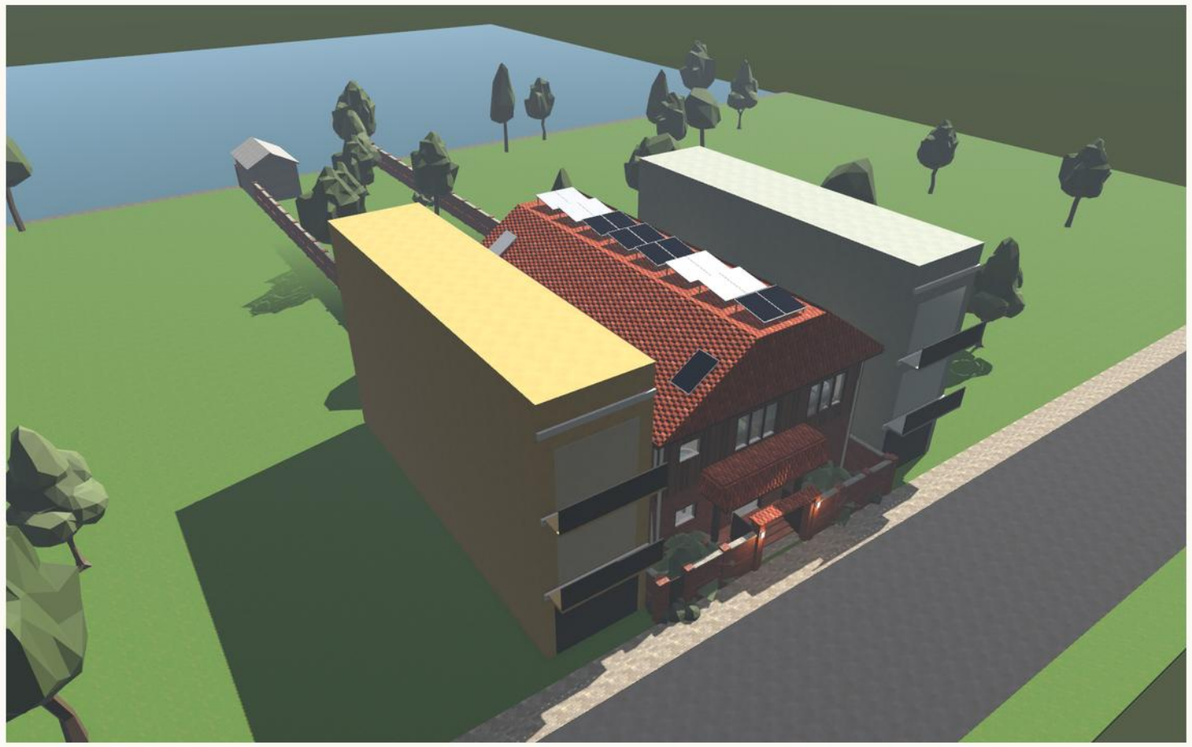
Interactive model: these are fixed viewpoints you can jump to with keys 1–9



Sun study: winter solstice, 12:00 solar time



Sun study: winter solstice, 15:00 solar time



Sun study: summer solstice, 09:00 solar time

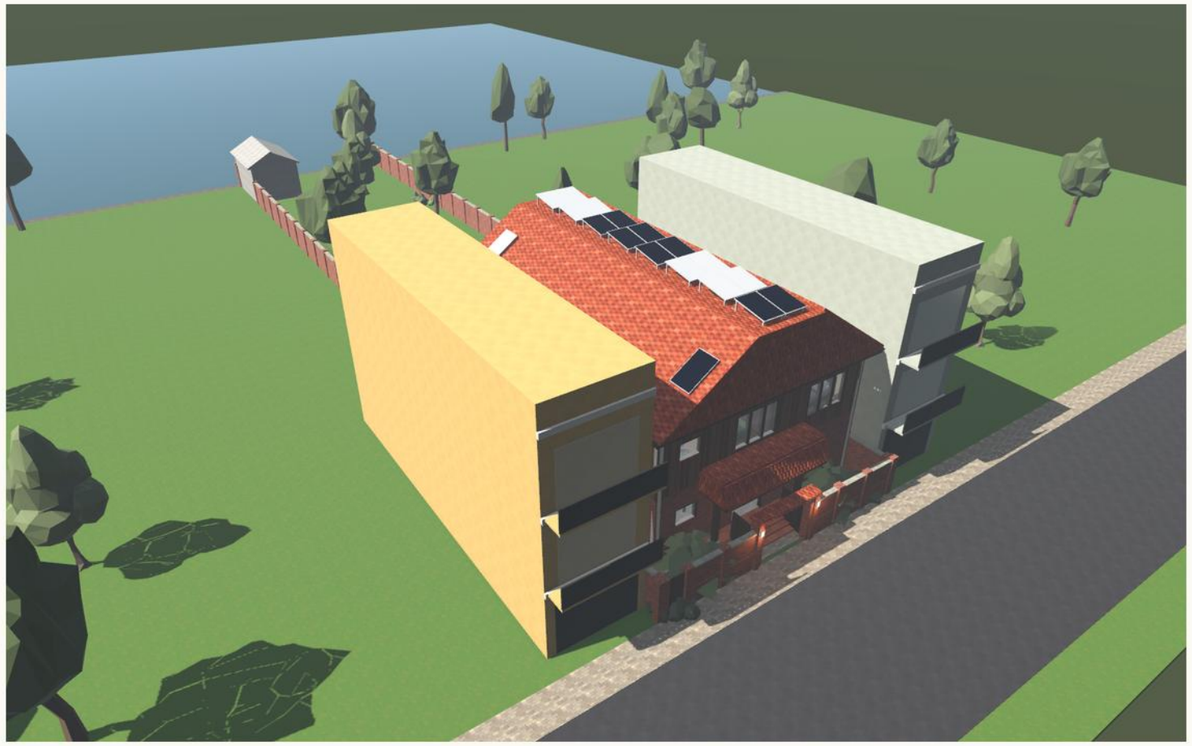


Sun study: summer solstice, 12:00 solar time

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Walkthrough (Unity)

Interactive model: these are fixed viewpoints you can jump to with keys 1–9



Sun study: summer solstice, 15:00 solar time

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits